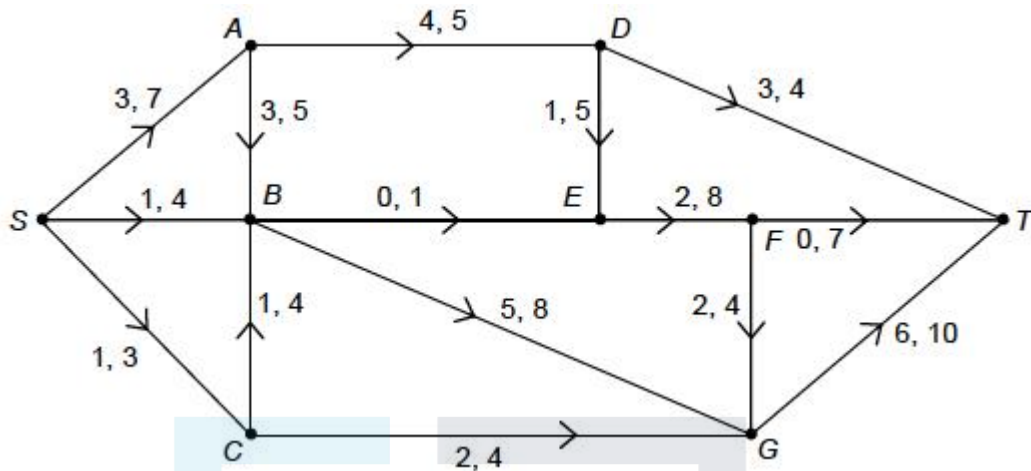


Q1.

The network shows a system of pipes, where S is the source and T is the sink.

The lower and upper capacities, in litres per second, of each pipe are shown on each arc.



- (a) There is a feasible flow from S to T .
- (i) Explain why arc AD must be at its lower capacity. (1)
- (ii) Explain why arc BE must be at its upper capacity. (1)
- (b) Explain why a flow of 11 litres per second through the network is impossible. (1)
- (c) The network in **Figure 1** shows a second system of pipes, where S is the source and T is the sink.

The lower and upper capacities, in litres per second, of each pipe are shown on each edge.

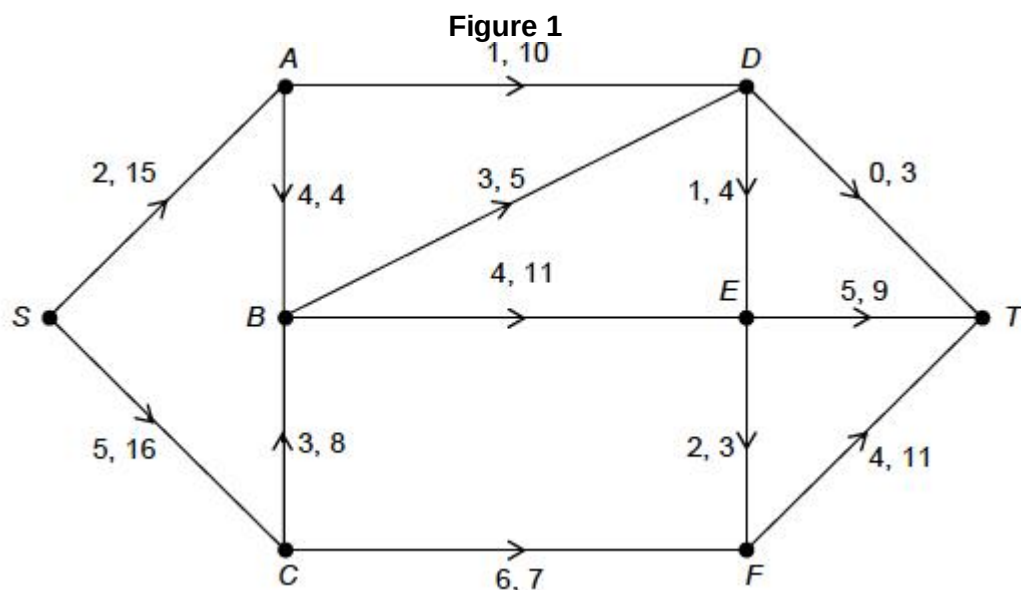
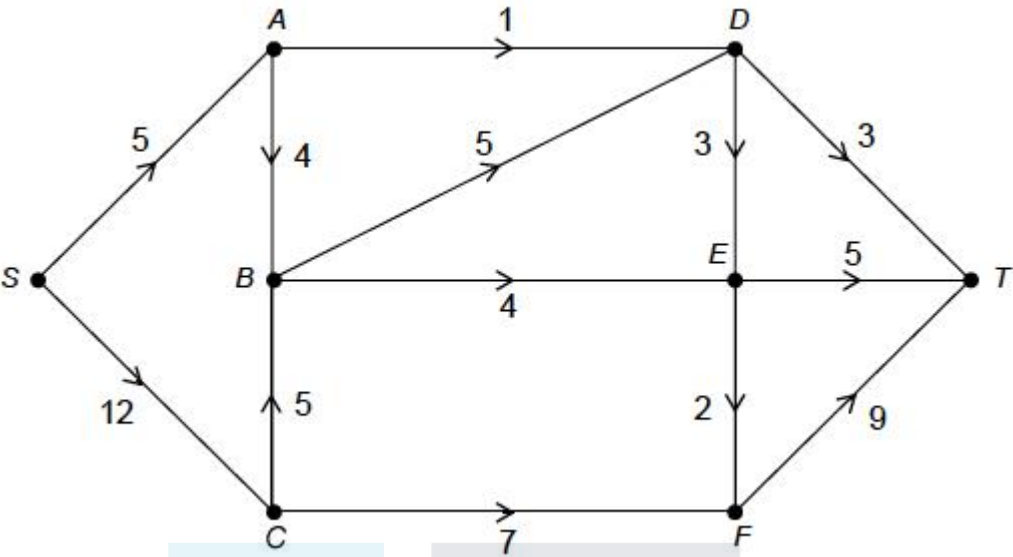


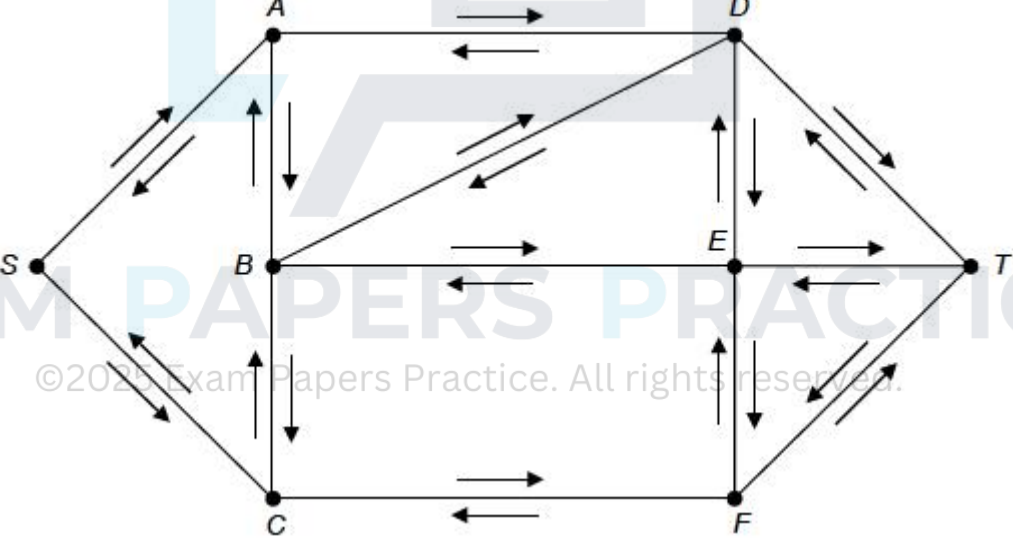
Figure 2 shows a feasible flow of 17 litres per second through the system of pipes.

Figure 2



- (i) Using **Figures 1** and **2**, indicate on **Figure 3** potential increases and decreases in the flow along each arc.

Figure 3



(2)

- (ii) Use flow augmentation on **Figure 3** to find the maximum flow from **S** to **T**.

You should indicate any flow augmenting paths clearly in the table below and modify the potential increases and decreases of the flow on **Figure 3**.

(1)

Augmenting Path	Flow



- (iii) Prove the flow found in part (c)(ii) is maximum.

(1)

- (iv) Due to maintenance work, the flow through node E is restricted to 9 litres per second.

Interpret the impact of this restriction on the maximum flow through the system of pipes.

(2)

(Total 11 marks)

Q2.

- (a) When investigating three network flow problems, a student finds:

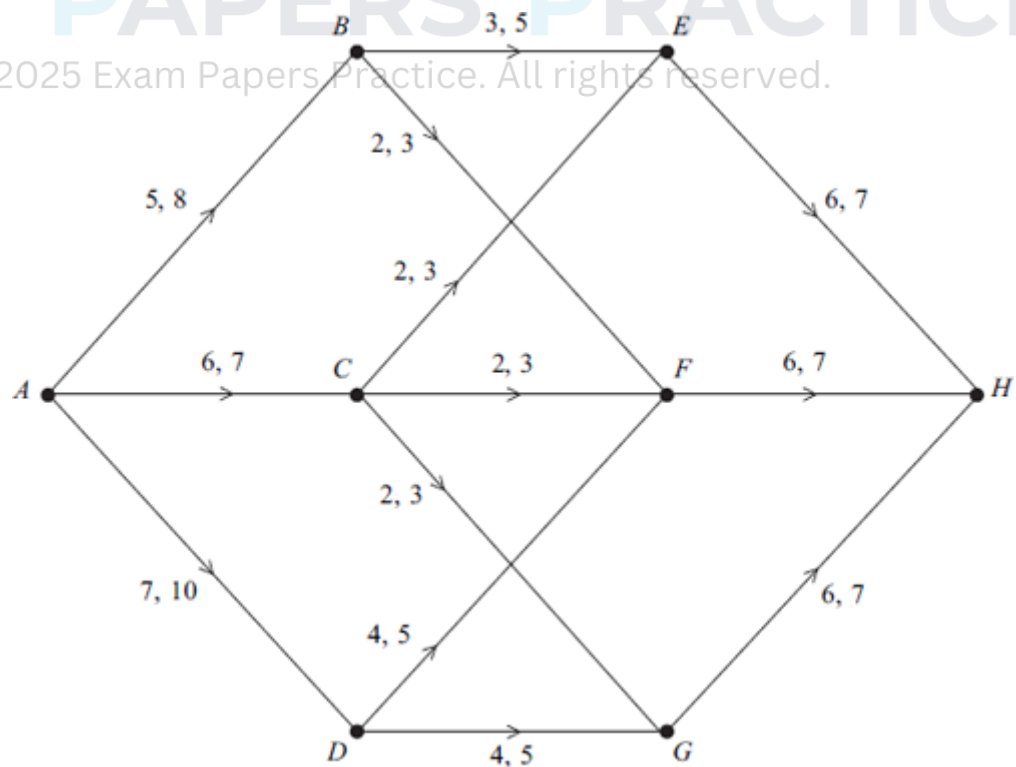
- (i) a flow of 50 and a cut with capacity 50;
- (ii) a flow of 35 and a cut with capacity 50;
- (iii) a flow of 50 and a cut with capacity 35.

In each case, write down what the student can deduce about the maximum flow.

(4)

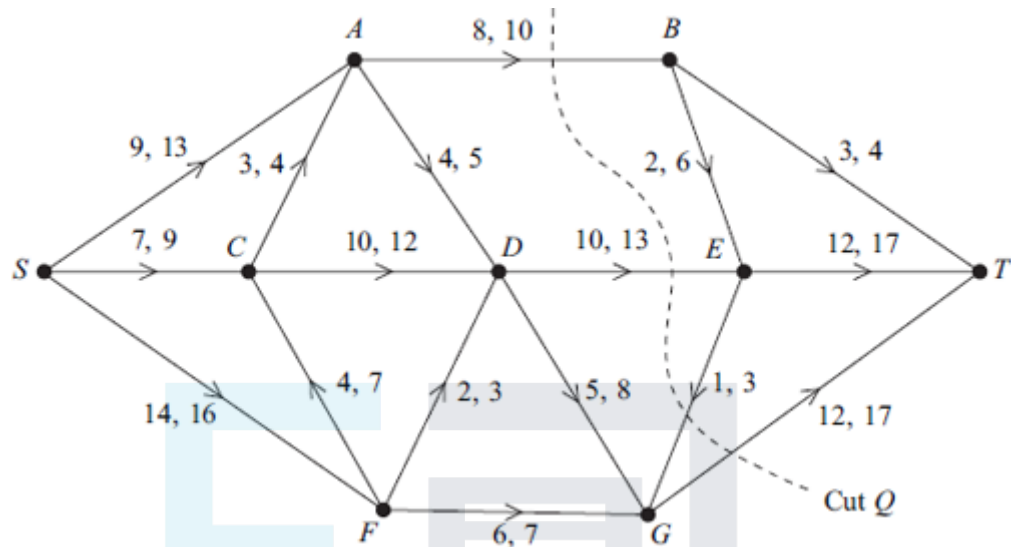
- (b) The diagram below shows a network. The numbers on the arcs represent the minimum and maximum flow along each arc respectively.

By considering the flow at an appropriate vertex, explain why a flow is not possible through this network.



Q3.

The network shows a system of pipes with the lower and upper capacities for each pipe in litres per second.



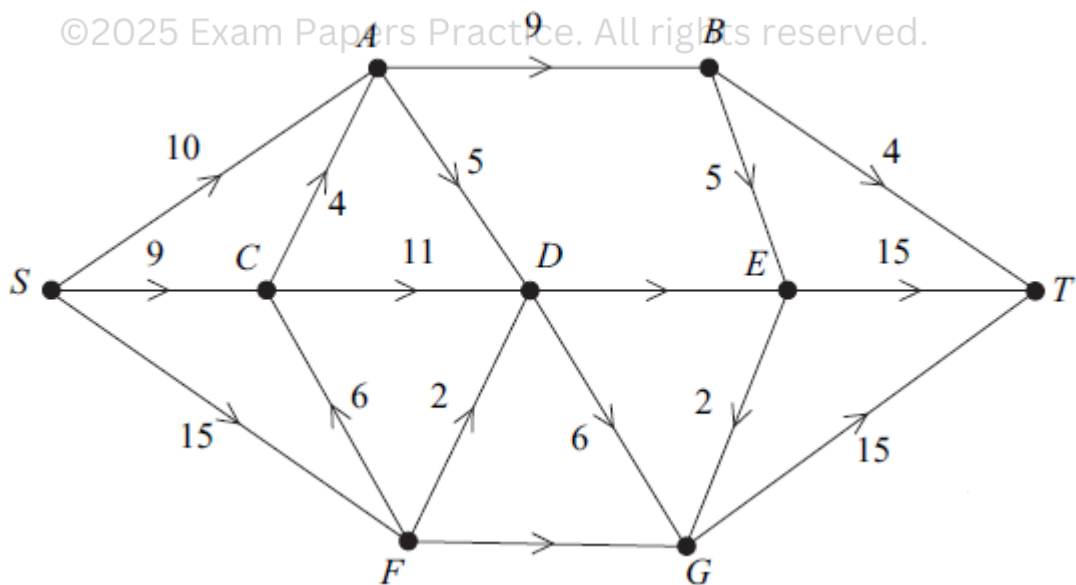
- (a) Find the value of the cut Q .

(2)

- (b) **Figure 1** shows most of the values of a feasible flow of 34 litres per second from S to T .

- (i) Insert the missing values of the flows along DE and FG on **Figure 1**.

Figure 1

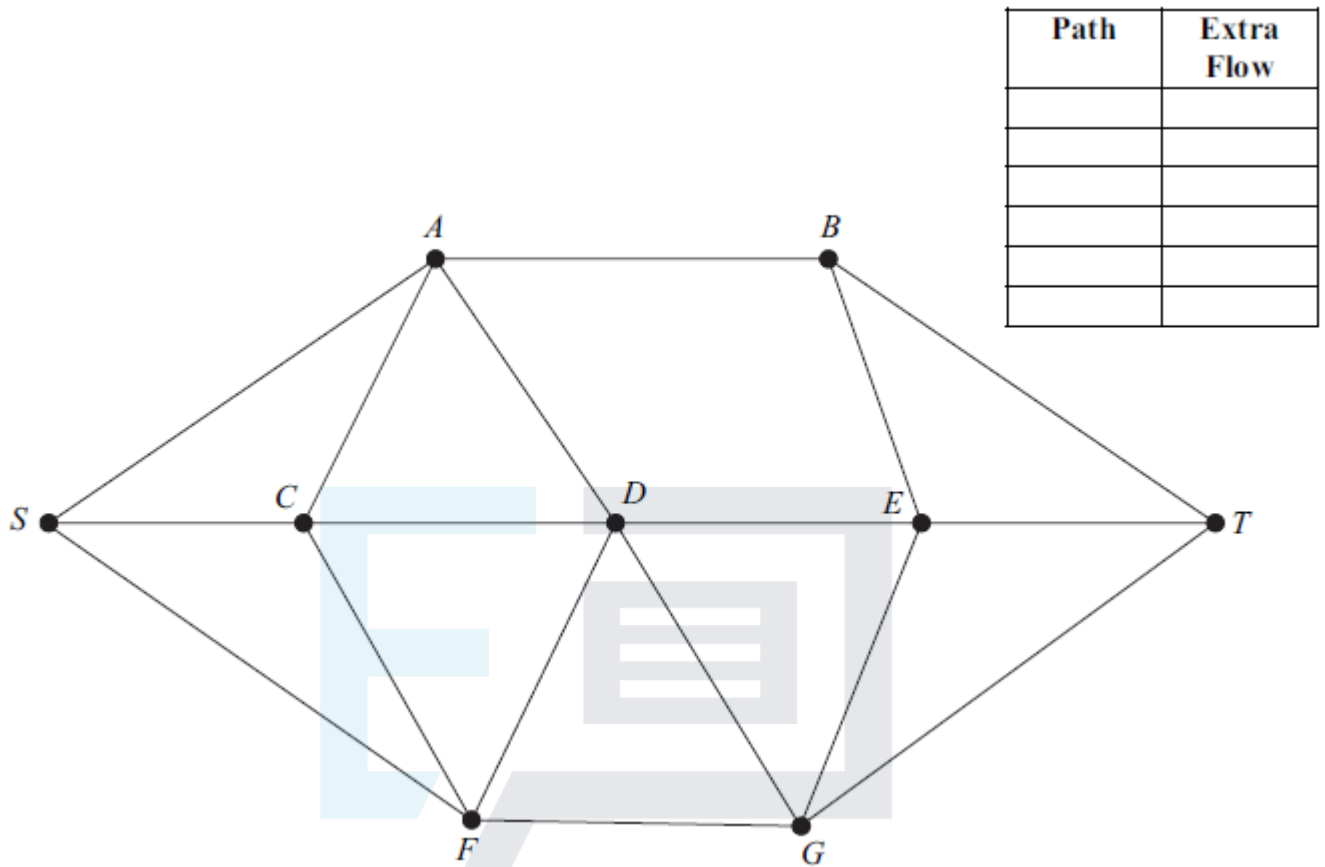


(2)

- (ii) Using this feasible flow as the initial flow, indicate potential increases and

decreases of the flow along each edge on **Figure 2**.

Figure 2



(2)

- (iii) Use flow augmentation on **Figure 2** to find the maximum flow from S to T . You should indicate any flow-augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

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(4)

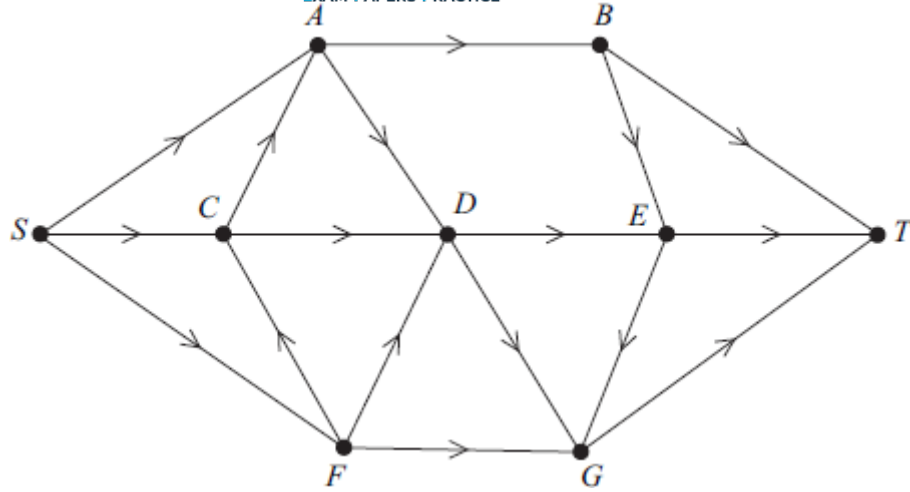
- (c) (i) State the value of the maximum flow.

Maximum flow is _____

(1)

- (ii) Illustrate your maximum flow on **Figure 3**.

Figure 3



(2)

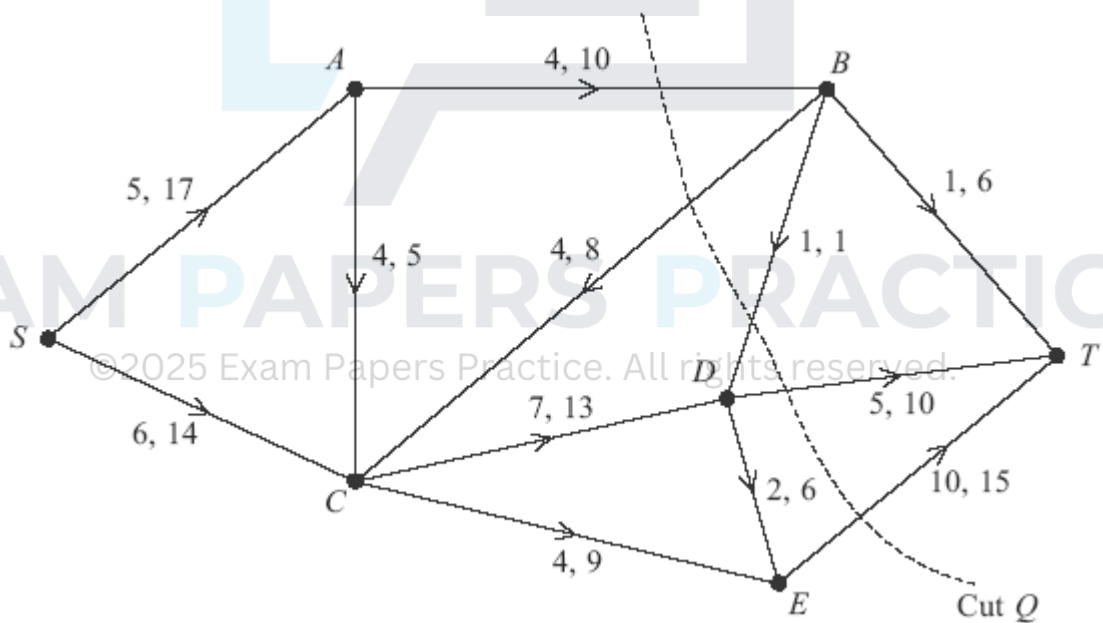
- (d) Find a cut with capacity equal to that of the maximum flow.

(1)

(Total 14 marks)

Q4.

The network shows a system of pipes with the lower and upper capacities for each pipe in litres per minute.

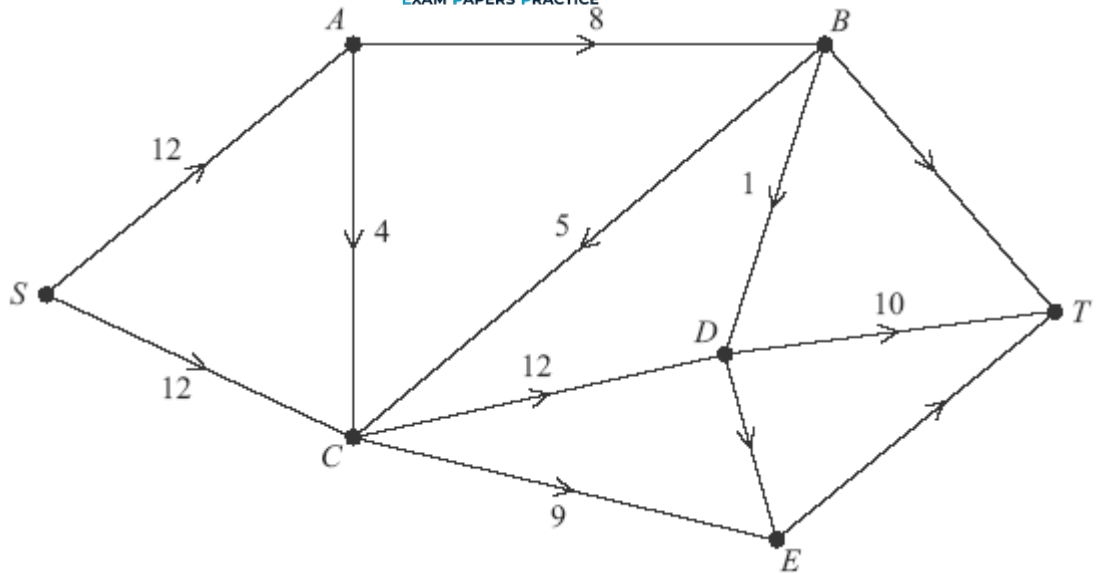


- (a) Find the value of the cut Q .

(2)

- (b) **Figure 1** below shows a partially completed diagram for a feasible flow of 24 litres per minute from S to T . Indicate, on **Figure 1**, the flows along the edges BT , DE and ET .

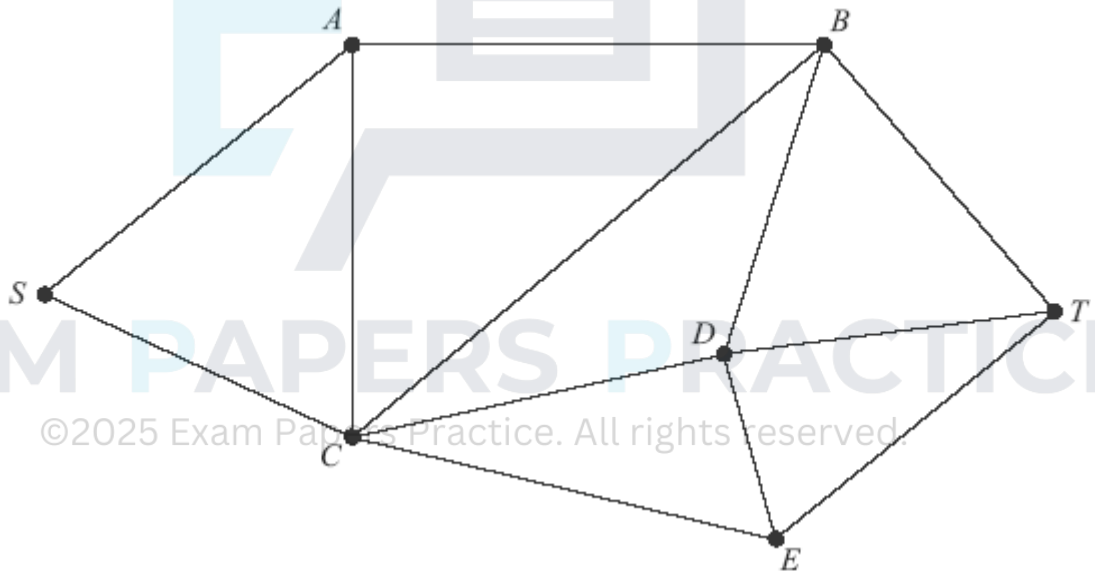
Figure 1



(2)

- (c) (i) Taking your answer from part (b) as an initial flow, indicate potential increases and decreases of the flow along each edge on **Figure 2** below.

Figure 2



Path	Flow

(2)

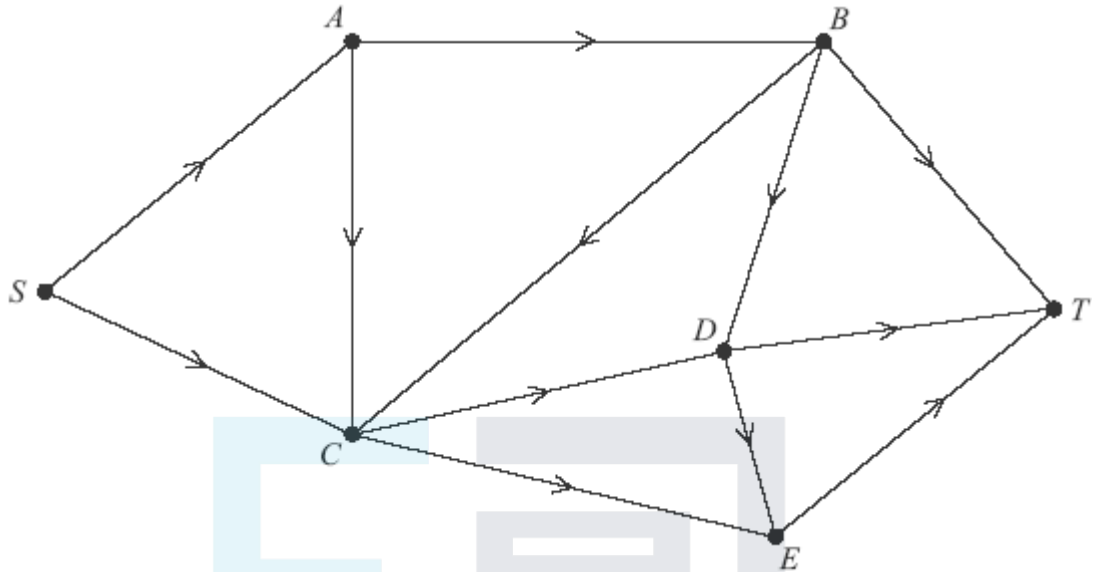
- (ii) Use flow augmentation on **Figure 2** to find the maximum flow from S to T .
You should indicate any flow augmenting paths in the table and modify the

potential increases and decreases of the flow on the network.

(5)

- (iii) Illustrate the maximum flow on **Figure 3** below.

Figure 3



(2)

- (d) Find a cut with value equal to that of the maximum flow.

You may wish to show the cut on the network above.

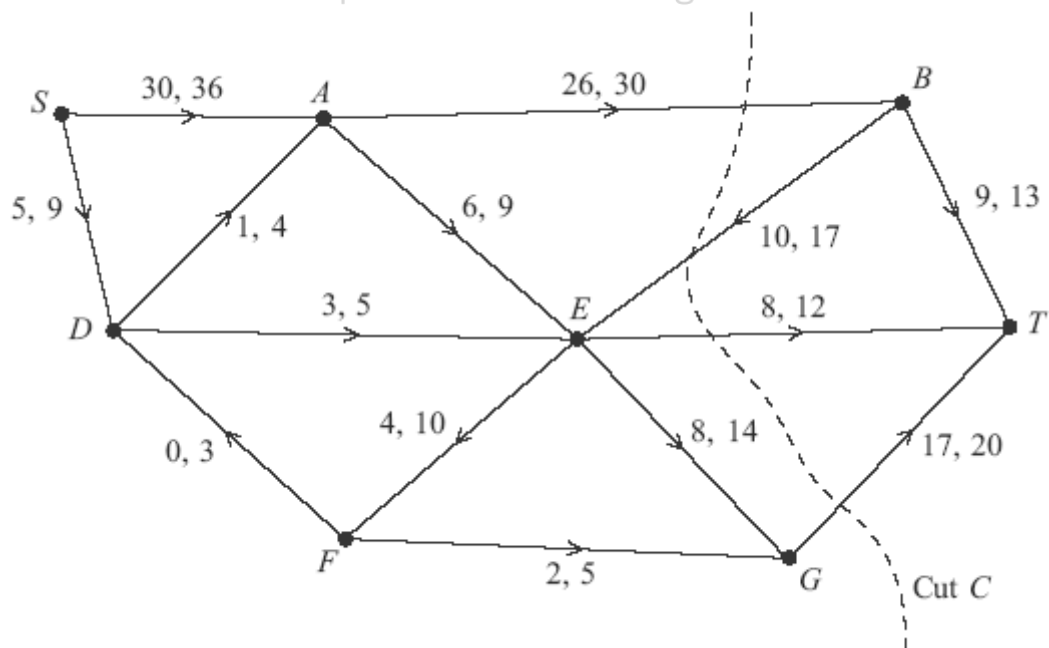
(1)

(Total 14 marks)

Q5.

The network shows a system of pipes with the lower and upper capacities for each pipe in litres per second.

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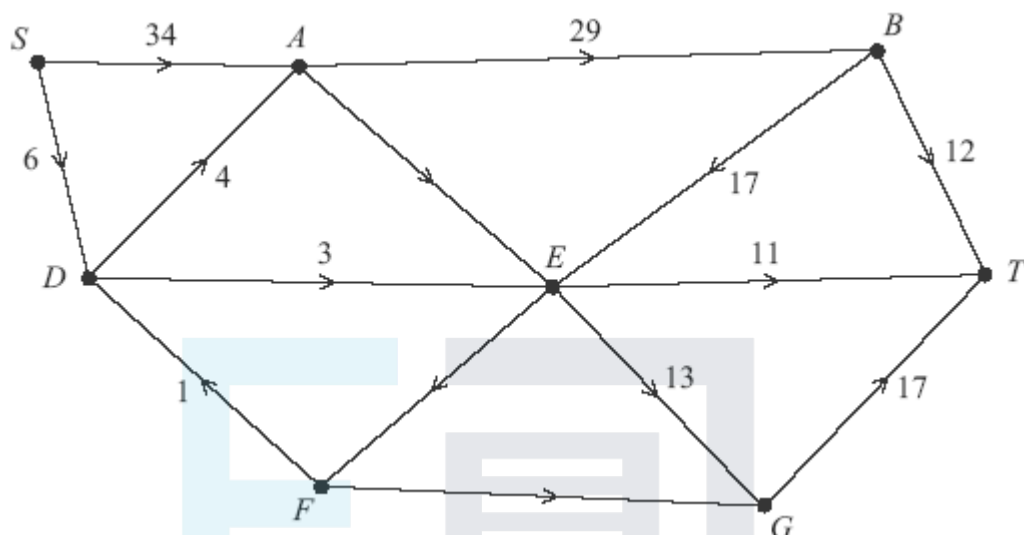


- (a) Find the value of the cut C .

(2)

- (b) **Figure 1**, below, shows a partially completed diagram for a feasible flow of 40 litres per second from S to T . Indicate, on **Figure 1**, the flows along the edges AE , EF and FG .

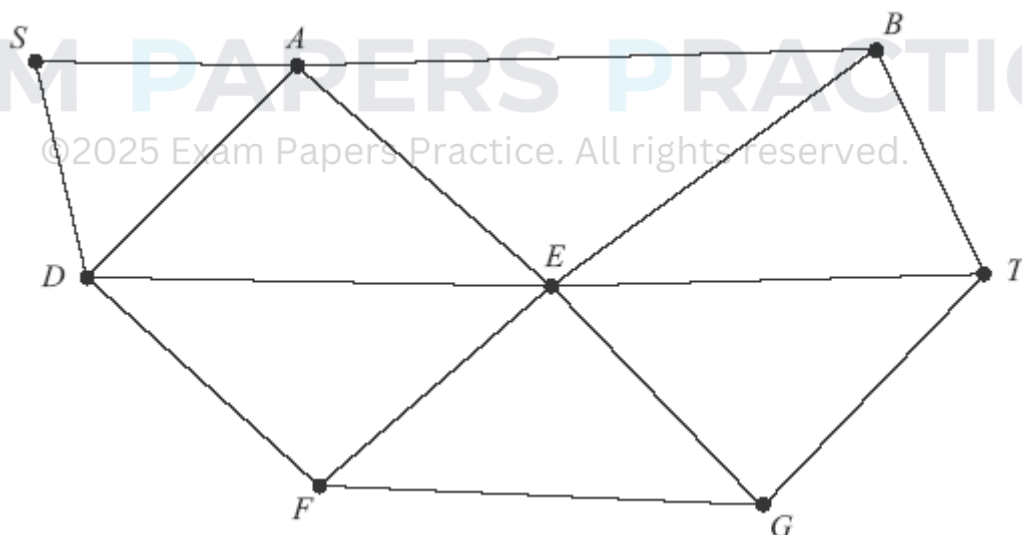
Figure 1



(3)

- (c) (i) Taking your answer from part (b) as an initial flow, indicate potential increases and decreases of the flow along each edge on **Figure 2**.

Figure 2



Path	Extra Flow

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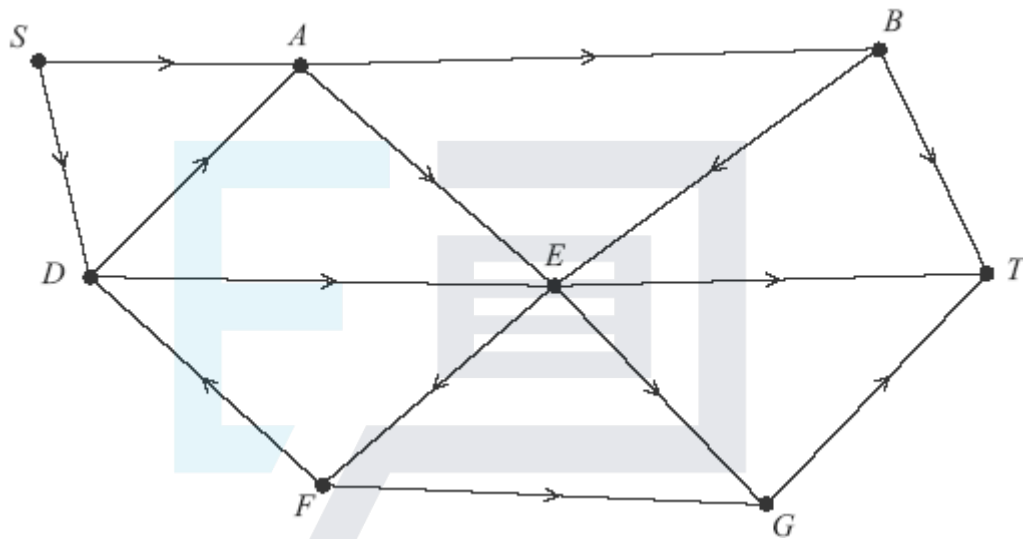
(3)

- (ii) Use flow augmentation on **Figure 2** to find the maximum flow from S to T . You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

(4)

- (d) Illustrate the maximum flow on **Figure 3**.

Figure 3



(2)

- (e) Find a cut with value equal to that of the maximum flow.

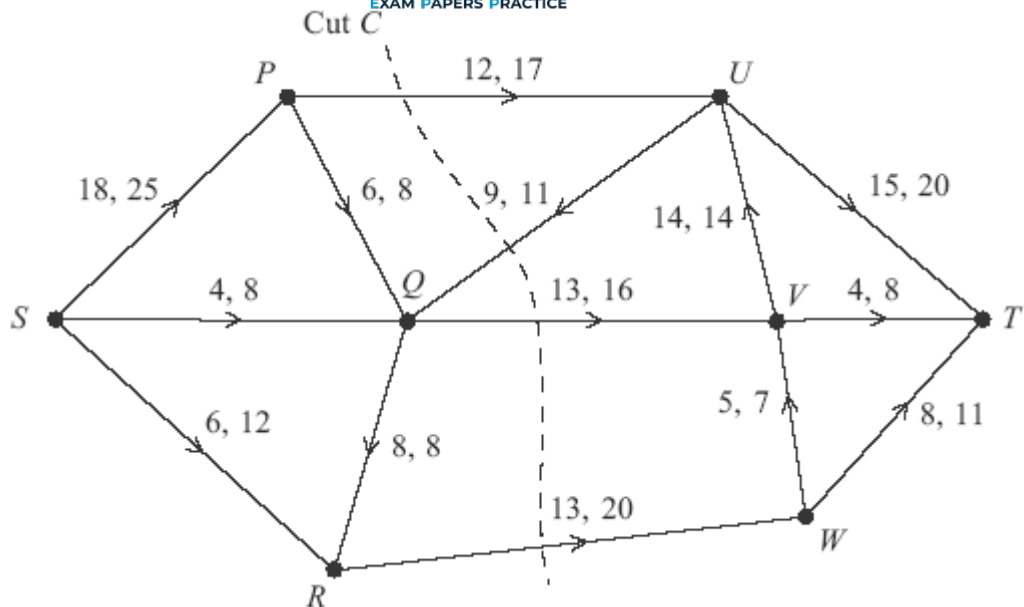
(2)

(Total 16 marks)

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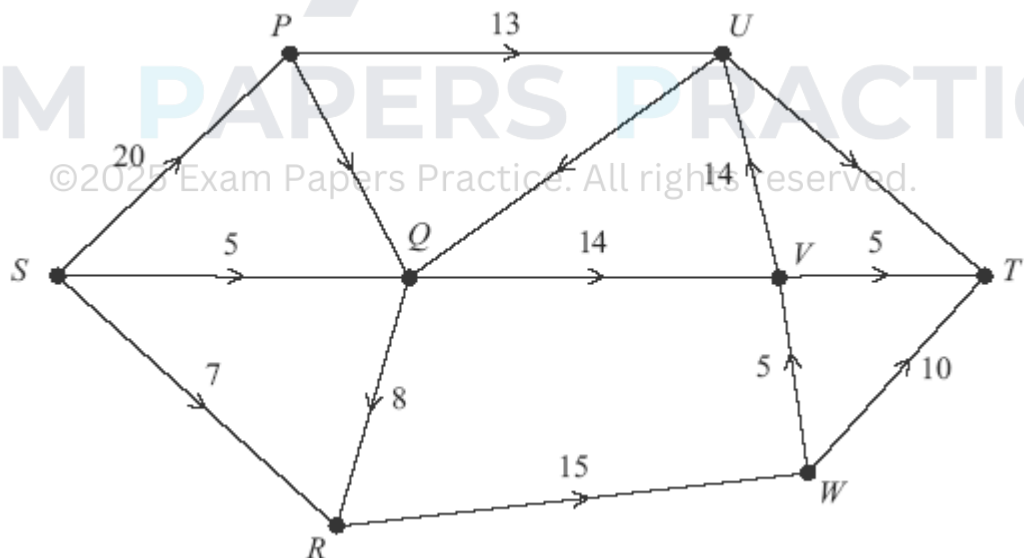
Q6.

The network shows a system of pipes with the lower and upper capacities for each pipe in litres per second.



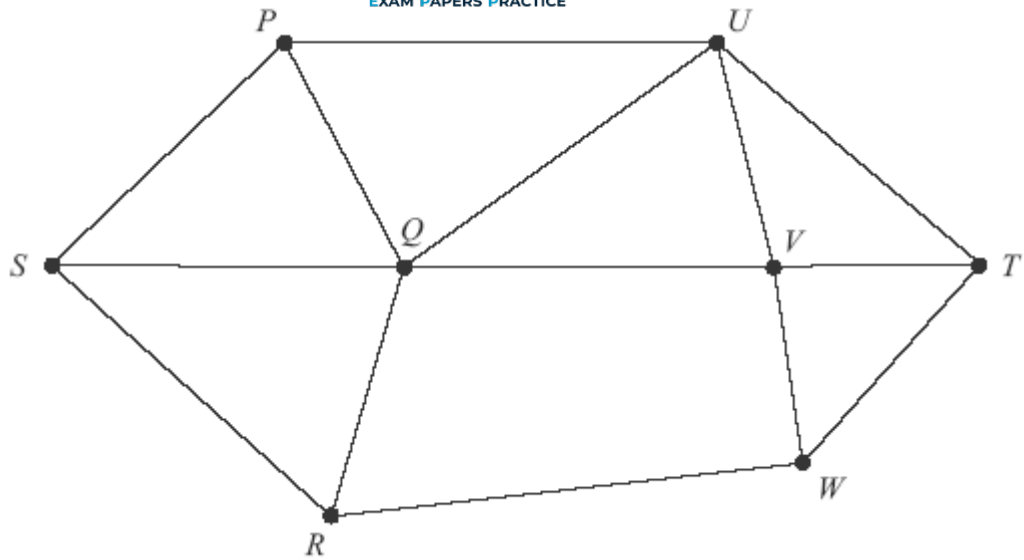
- (a) (i) Find the value of the cut C . (1)
- (ii) Hence state what can be deduced about the maximum flow from S to T . (1)
- (b) **Figure 1**, below, shows a partially completed diagram for a feasible flow of 32 litres per second from S to T . Indicate, on **Figure 1**, the flows along the edges PQ , UQ and UT .

Figure 1



- (c) (i) Taking your feasible flow from part (b) as an initial flow, indicate potential increases and decreases of the flow along each edge on **Figure 2**. (3)

Figure 2



Path	Additional Flow

(2)

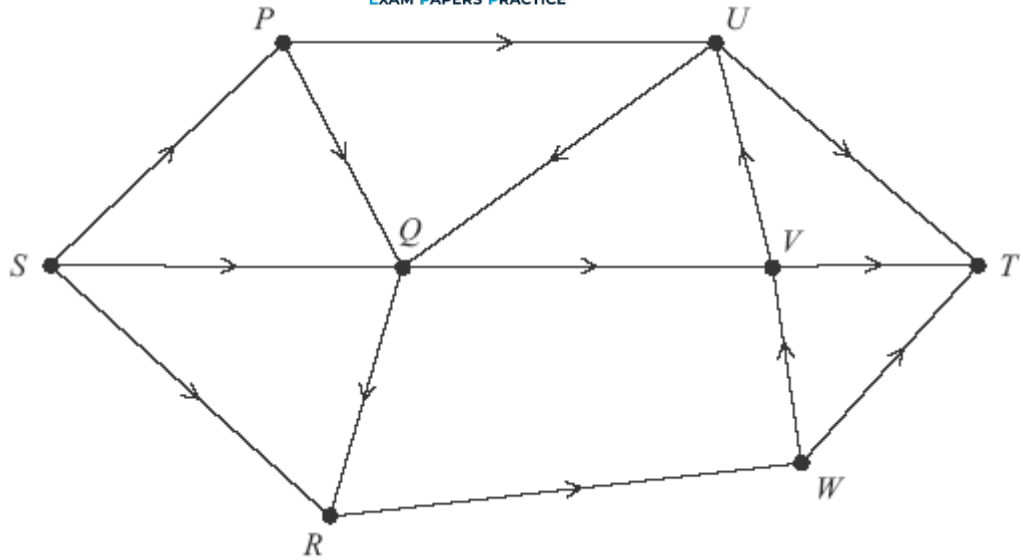
- (ii) Use flow augmentation on **Figure 2** to find the maximum flow from S to T . You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

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(5)

- (iii) Illustrate the maximum flow on **Figure 3**.

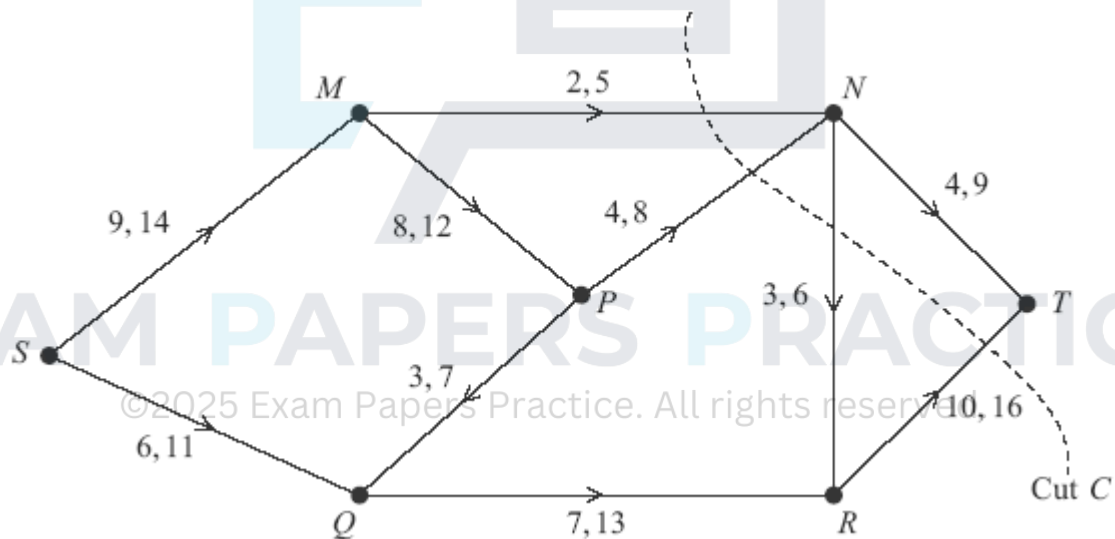
Figure 3



(1)
(Total 13 marks)

Q7.

The network shows a system of pipes with the lower and upper capacities for each pipe in litres per second.



(a) (i) Find the value of the cut C .

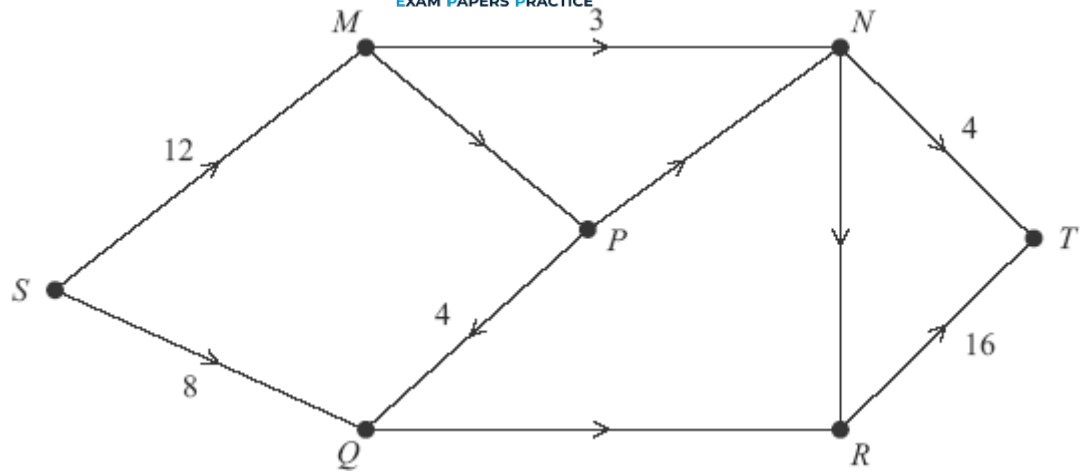
(1)

(ii) State what can be deduced about the maximum flow from S to T .

(1)

(b) **Figure 1** shows a partially completed diagram for a feasible flow of 20 litres per second from S to T . Indicate, on **Figure 1**, the flows along the edges MP , PN , QR and NR .

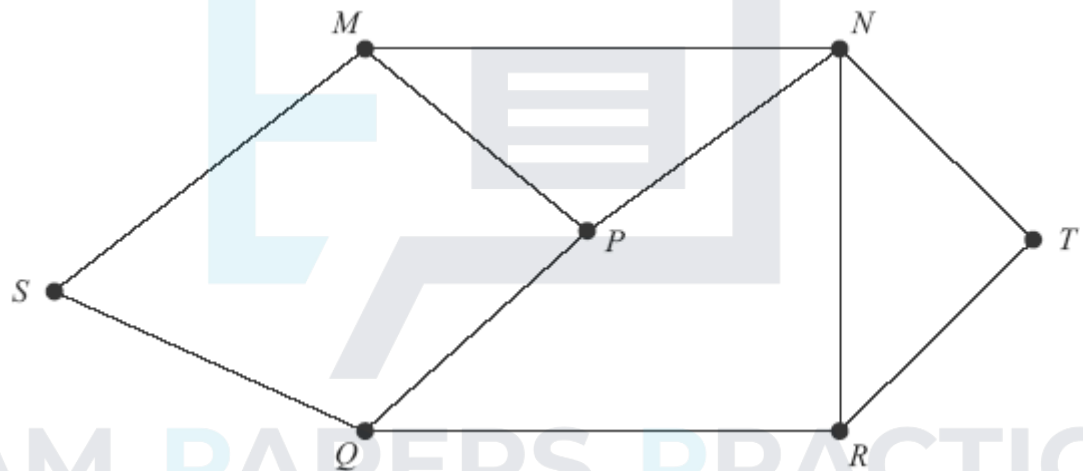
Figure 1



(4)

- (c) (i) Taking your answer from part (b) as an initial flow, indicate potential increases and decreases of the flow along each edge on **Figure 2**.

Figure 2



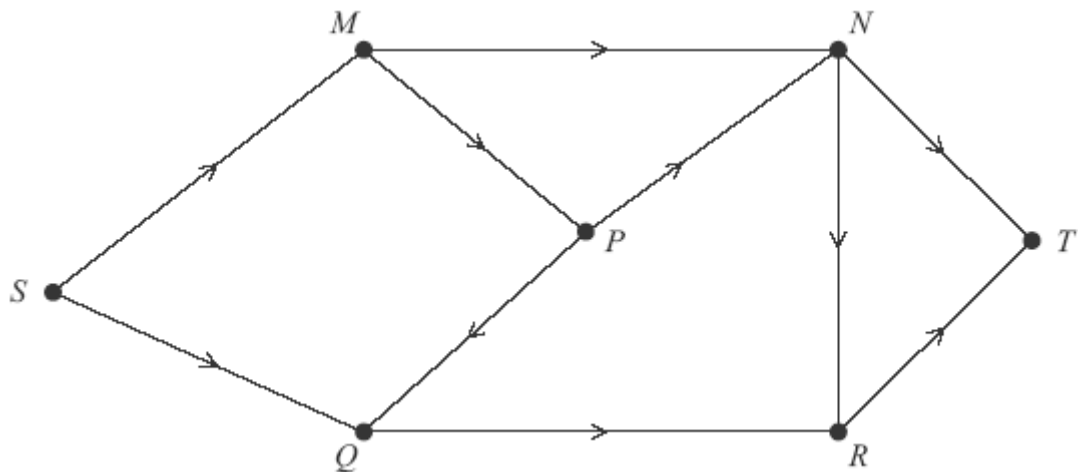
Path	Flow

(2)

- (ii) Use flow augmentation on **Figure 2** to find the maximum flow from S to T . You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

(5)

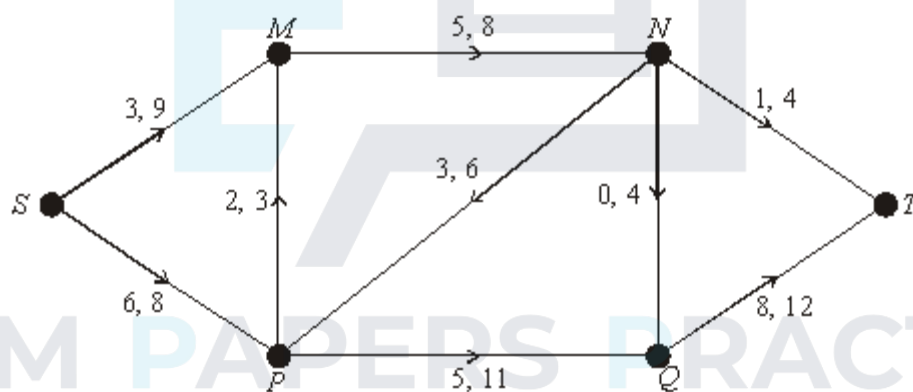
- (iii) Illustrate the maximum flow on **Figure 3**.



(2)
(Total 15 marks)

Q8.

The network shows a system of pipes, with the lower and upper capacities for each pipe in litres per second.



- (a) **Figure 1**, below, shows a partially completed diagram for a feasible flow of 10 litres per second from S to T . Indicate, on **Figure 1**, the flows along the edges MN , PQ , NP and NT .

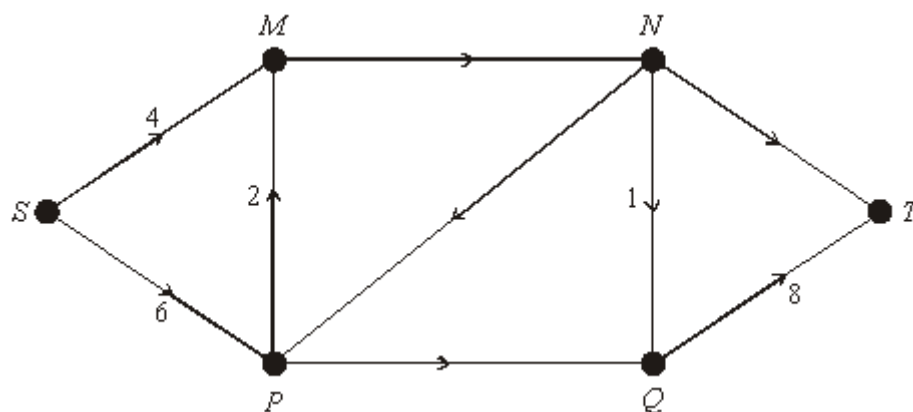


Figure 1

(4)

- (b) (i) Taking your answer from part (a) as an initial flow, use flow augmentation on **Figure 2** to find the maximum flow from S to T .

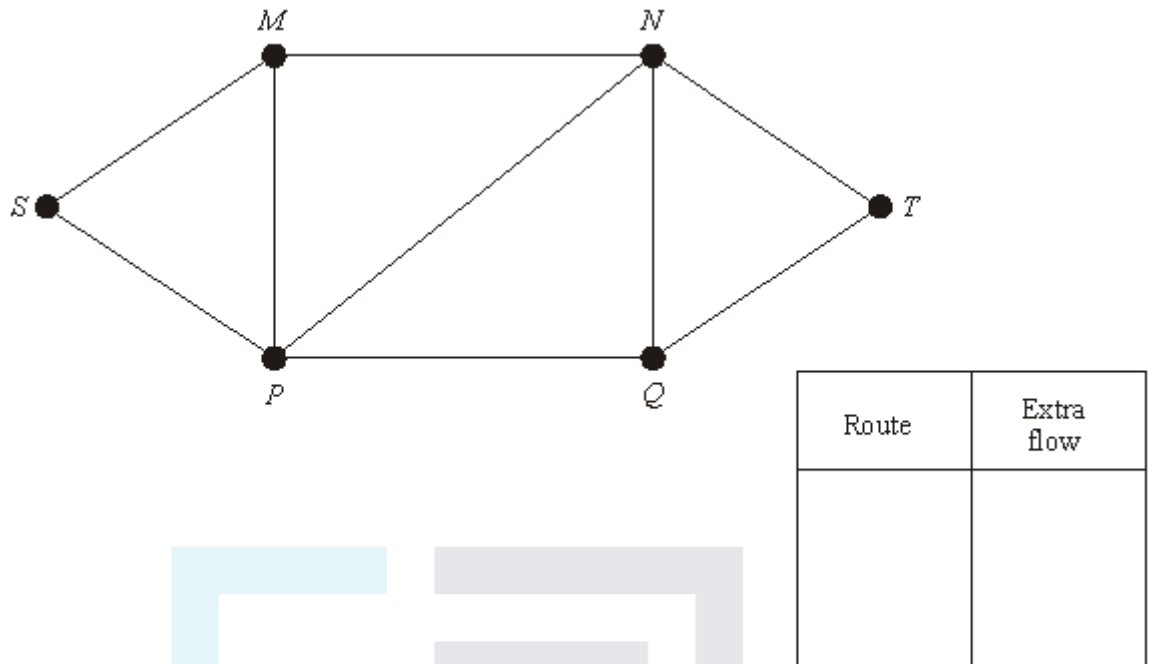


Figure 2

(6)

- (ii) State the value of the maximum flow and illustrate this flow on **Figure 3**.

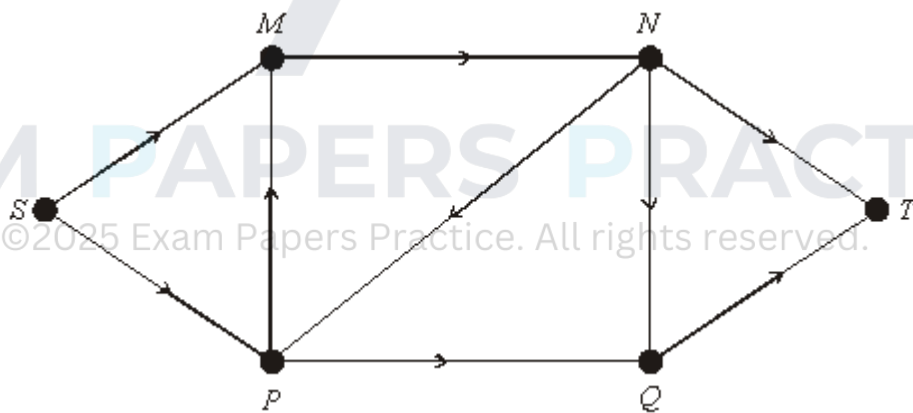


Figure 3

(2)

- (c) Find a cut with capacity equal to that of the maximum flow.

(2)

(Total 14 marks)

Q9.

- (a) When investigating a network flow problem, a student finds:
- (i) a cut with capacity 25 and a flow of 25;

- (ii) a cut with capacity 25 and a flow of 15.

In each case, write down what the student can deduce.

(2)

- (b) **Diagram 1**, below, shows a network flow problem. The numbers on the arcs represent the minimum and maximum flow along each arc respectively. By considering the flow at an appropriate vertex, explain why a flow is **not** possible through this network.

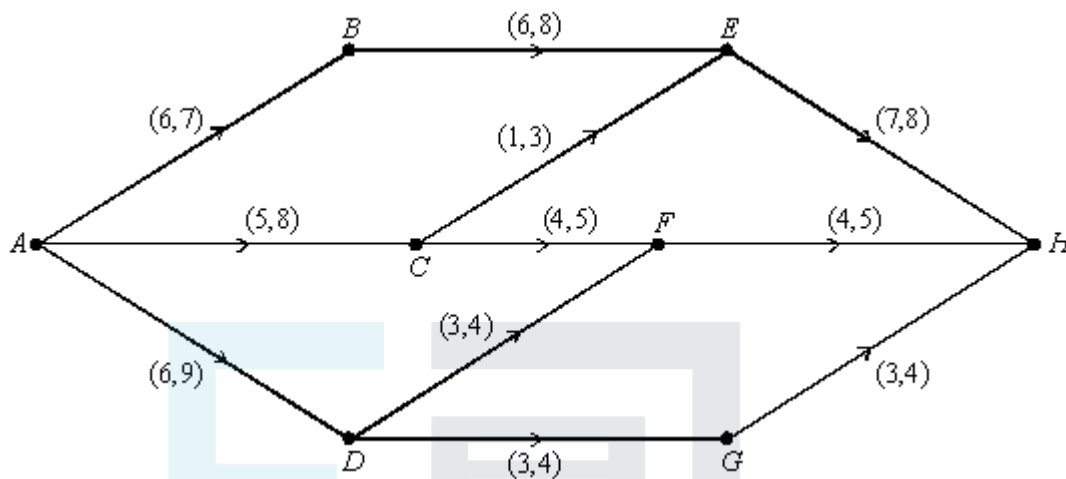


Diagram 1

(2)

- (c) **Diagram 2**, below, shows the maximum flow along each arc in a network flow problem.

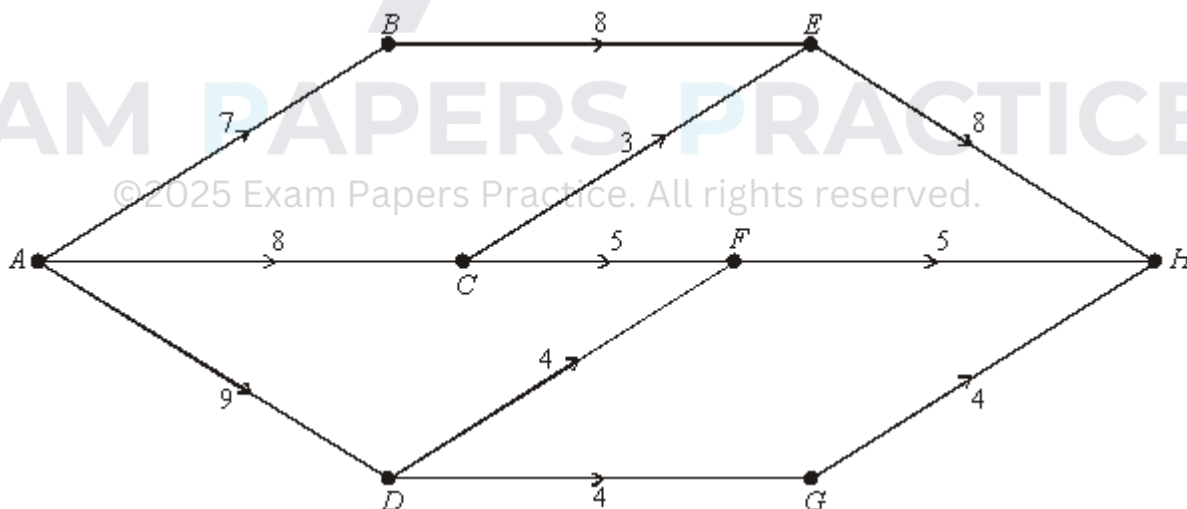


Diagram 2

- (i) On **Diagram 2**, starting from a position of zero flow, use flow augmentation to find the maximum flow.

(5)

- (ii) Confirm that you have a maximum flow by finding a cut of the same value.

(1)



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